Computational Imaging Group B Project Template

Group B
Number of students: 2

1. General Description

In this project, students are asked to address a computational imaging problem by combining different methodological families, in order to understand both classical and modern approaches and to critically compare their behavior under the same experimental conditions. The goal is not only to obtain good reconstruction results, but also to reason about the strengths and limitations of each method.

2. Dataset

Dataset to be used: Mayo
Download link:
https://drive.google.com/drive/folders/13BEiz6t57qSbwBpCtfqllmYTLmkhQeFE?usp=drive_link

Students are expected to carefully inspect the dataset, understand its structure, and define a consistent preprocessing pipeline. This includes normalization, resizing if needed, and splitting into training, validation, and testing sets when applicable. Any preprocessing choice must be clearly justified and documented.

3. Task Definition

Target task: Sparse-views CT

The images must be resized to 256x256.

The task must be formulated as an inverse problem where a degraded observation is mapped back to a high-quality reconstruction.

Parameters of geometry:
number of angles in (180,90,60,45)

noise level = 0.005

Students must ensure all methods use the same degraded inputs.

4. Methodological Requirements

The students are expected to test and compare multiple methods on the same data. In particular:

OBBLIGATORIO **Variational method:** regularization with Total p-Variation (TpV) with p in (0.1,0.5)

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End-to-end method: Select one appropriate method within the following list of architecutres: UNet, ViT, or NAF-Net. Choose specific architecture and training in relation to the task properties, by using the ground truth images as target.
Generative method: Diffusion models with DiffPir suitably adapted to this task, explain usage.
~~**Hybrid method**~~**:** PD-Net with TV regularization with a number of iterations to unroll defined by the student, describe integration. SCARTATO

If the group is constituted by only 2 students, they can choose three of the previous four methods, by excluding one of the last three (they can also do all the four, obviously!).
Each of the previous methods has it own set of parameters that needs to be chosen, e.g. the number of diffusion steps for DPS or the regularization parameter. The students are expected to identify the values of such parameters heuristically and discuss the final choice, together with a description of how they set them, at the oral presentation.
5. Deliverables

During the oral presentation, the students are expected to provide a comparison of the tested methods in term of the following:

Quantitative: PSNR, SSIM.
Visual: reconstructed images and comparisons.

In particular, they should include a comparison plot within all methods, a report with methods, setup, and discussion.
Finally, we expect the student to provide a code with documentation, possibly pushed on Github whose link should be delivered during the oral presentation.
During the oral presentation, the students should have a PowerPoint / Beamer presentation reporting all the deliverables and the discussion as indicated in the project description.